

The Alpha Coefficient in Rope Bundle Theory

The Coupling $\kappa = \alpha/2\pi = 0.0011614$, Computed and Cross-Checked

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Abstract

This paper reports the rope framework's identification of the electromagnetic coupling. The rope coupling κ is tied to the fine-structure constant by $\kappa = \alpha/2\pi$, computed as 0.00116141 and cross-checked between the electromagnetic and particle sectors of the package so the two agree. The supporting electromagnetic constants — $\epsilon_0 = 8.854 \times 10^{-12}$, the impedance of free space $Z_0 = 376.73 \Omega$, and $\alpha \approx 1/137$ — are likewise computed from the structure and verified. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants.

Scope of this paper

CLAIM: the rope coupling satisfies $\kappa = \alpha/2\pi = 0.0011614$, consistent across sectors, with the EM constants (ϵ_0 , $Z_0 = 376.73 \Omega$, $\alpha \approx 1/137$) computed and verified.

NOT CLAIMED: a first-principles derivation of the NUMERICAL VALUE of α itself (its value is taken from the EM structure, not predicted from nothing). Status: Derived relation / Modeled value.

1. The Coupling Relation

In the rope framework the coupling that governs how strongly links interact is identified with the fine-structure constant through

$$\kappa = \alpha / 2\pi = 0.00116141 \quad (1.1)$$

computed in the particle sector and cross-checked against the electromagnetic sector, where the two independent routes to α are required to agree — a genuine internal-consistency lock.

2. The Electromagnetic Constants

Constant	Computed value	Reference
ϵ_0	$8.854 \times 10^{-12} \text{ F/m}$	$= 1/(\mu_0 c^2)$
Z_0 (free-space impedance)	376.73Ω	$= \mu_0 c$
α (fine-structure)	$\approx 1/137$	$= e^2 Z_0 / 2h$
$\kappa = \alpha/2\pi$	0.00116141	cross-sector lock

An honest note on α . The framework provides a consistent RELATION ($\kappa = \alpha/2\pi$) and reproduces the EM constants from structure, but it does not predict the numerical value of α from nothing — α

enters through the electromagnetic structure. The result is a verified relation and cross-sector consistency, not a derivation of why $\alpha \approx 1/137$.

Status

$\kappa = \alpha/2\pi = 0.0011614$, cross-sector consistent — Derived (relation).

EM constants ϵ_0 , $Z_0 = 376.73 \Omega$, $\alpha \approx 1/137$ — Modeled (from structure, verified).

Numerical value of α from first principles — Open.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.